

Phase 5.1 Production Hardening Report

Appliance Baseline: RCS CyberTrack Core Management Backend | **Status:** 100% PASSED (88/88 Tests)

EXECUTIVE SUMMARY:

Phase 5.1 Production Hardening has been successfully implemented across all core subsystems of RCS CyberTrack. All 12 requested production hardening tasks have been fulfilled and verified with zero regressions. The test suite expanded from 77 baseline tests to **88 passed tests** (100% pass rate across 7 test modules).

1. Implemented Hardening Measures

- **RBAC Permission Boundaries:** Hardened endpoint dependency checks in `server.py` to enforce exact domain permissions (`firewall.read/write`, `network.read/write`, `alerts.write`, `system.write`, `vpn.read/write`, `sdwan.read`, `analytics.read`).
- **User Self-Protection & Last-Admin Safeguards:** Enforced strict checks in `user_service.py` blocking active administrators from disabling their own account or altering their role level. Enforced safeguards preventing demotion or disabling of the sole active administrator.
- **Configuration Engine Lifecycle & Verified Rollback:** Hardened candidate commit state machine in `config_engine.py`. Implemented post-apply health verification in `os_adapter.py`. Guaranteed post-rollback health verification for automatic and manual rollbacks prior to reporting success.
- **nftables IPv4/IPv6 & Input Sanitization:** Updated `compiler.py` to compile IPv4 and IPv6 rules dynamically (`ip saddr` vs `ip6 saddr`), rejecting mixed family rules. Hardened regex to reject newlines, null bytes, semicolons, shell metacharacters, and command strings.
- **Configuration-Driven Management Ports:** Made API (default 8000) and SSH (default 22) management ports configurable via `PolicyConfig` while preserving full backward compatibility.
- **Audit Log Credential Redaction:** Implemented recursive data sanitization in `audit.py`, automatically redacting sensitive keys (`password`, `secret`, `token`, `api_key`, `credential`) to `[REDACTED]`.
- **Hardening Test Suite & Documentation:** Created `tests/test_phase54_hardening.py` containing 11 comprehensive unit/integration tests and updated system documentation (`rbac.md`, `config-engine.md`, `nftables-security.md`).

2. Test Verification & Baseline Results

Test Module	Status	Test Description / Scope
test_api.py	PASSED	API authentication, system status, & RBAC rules
test_firewall.py	PASSED	Firewall rule management & Pydantic validation
test_network.py	PASSED	Network interface & routing table management
test_phase4b_api.py	PASSED	Phase 4b security alert & audit log endpoints
test_phase51_rbac_users.py	PASSED	Phase 5.1 user management & lockout protection
test_phase52_config_engine.py	PASSED	Phase 5.2 configuration staging, commit & diff
test_phase53_nftables.py	PASSED	Phase 5.3 nftables compiler & non-interactive sudo
test_phase54_hardening.py	PASSED	Phase 5.1 production hardening & security boundary tests
TOTAL SUITE RESULT	88 PASSED	100% Pass Rate (88 Passed in 31.15s)

3. Modified Repository Files Manifest

File Path	Status	Description of Modifications
docs/config-engine.md	Modified	Documented verified rollback guarantee & state machine
docs/nftables-security.md	Modified	Documented IPv4/IPv6, injection rejection, & mgmt ports
docs/rbac.md	Modified	Documented user self-protection & last-admin safeguards
firewall/compiler.py	Modified	IPv4/IPv6 support, injection regex, configurable ports
management/adapters/os_adapter.py	Modified	Health verification via fw_backend.verify()
management/api/server.py	Modified	Fine-grained domain permission dependencies on endpoints
management/audit/audit.py	Modified	Recursive credential & sensitive data redaction filter
management/services/config_engine.py	Modified	Strict commit state checks & verified rollback flow
management/services/user_service.py	Modified	Self-protection & last-admin protection enforcement
tests/test_phase54_hardening.py	CREATED	New test suite with 11 production hardening tests

4. Sign-Off & Verification Conclusion

Phase 5.1 Production Hardening is complete, verified, and ready for production deployment. All safety checks, process isolation rules, input validation routines, and regression tests meet high-assurance security standards.